

AUX–HV (builds PV, DAB–D60)

Rev C1 | 2026–10–04 | Schematics authored by Chinmoy Bhuyan | 22.8 V / 30 W isolated control rail from the DC ports, 200–1000 V (1100 V transient)

01 HV input, limit, fuse, OR

Per port: studs, 2 x 22 R pulse–rated AHEAD of the 1 A / 1000 VDC fuse, BYG10V OR of both poles; 2 x 3.3 uF / 1300 V film = surge absorber.
Port B is not fitted on the DAB–D60 build (bom/AUX–HV_DAB_BOM.csv).

File: sheets/01_input.kicad_sch

02 Start–up, brown–in, OV, timer

3 x BSS126 cascode on a compensated tap string; TPS3710 brown–in pulls COMP low; TLV3202 ch 1 = input OV lockout (COMP low),
ch 2 = latched overload timer (soft–start crowbar) –> hiccup

File: sheets/02_startup.kicad_sch

03 PWM, SiC, clamp, T1, foldback

UCC28C59–Q1 at 65 kHz, IV2Q171R0D7Z 1700 V SiC, TVS clamp, T1 (CUSTOM EC34A), aux –> BC817 follower –> VDD, aux backup loop on FB,
foldback: 10 nF onto RT/CT below –8 V output (no current staircase)

File: sheets/03_power.kicad_sch

04 Rectifier, output, feedback, PG

VS–8ETU04S + RC snubber, 3 x 100 uF polymer, ATL431 + CNY65B fast–lane type II with R–C feed–forward, TPS3700 window –> SN74LVC1G17 –> PG_HVAUX (active high), output connector (interfaces.AUX)

File: sheets/04_secondary.kicad_sch

BUILD OPTIONS: PV = all parts fitted (bom/AUX–HV_BOM.csv). DAB–D60 = port 1 only (D–021): the port–B input parts marked ‘PV build only’ (2 studs, 2 x 22 R, fuse, 2 OR diodes) are not fitted (bom/AUX–HV_DAB_BOM.csv).

CONFORMAL COATING (IC–17, D–032): coated to pollution degree 1 (IEC 60664–3 type 1) over the HV–PELV barrier parts T1, CNY65B and their HV copper. T1: PD <= 10 pC at 2461 Vpk on every unit, 4400 V rms 60 s, 8 kV impulse.

CALCULATED (sim/aux_hv_design.py rev C1): DCM flyback 64.9 kHz; start 228 V (worst 232 V), stop 175–190 V; out 22.40–23.15 V; eta 80.8/84.0/83.5/82.2/80.4 % at 200–1000 V (T1 design M1); switch IV2Q171R0D7Z (alt. SG2M1K0170J2J), gate 15.5–18.1 V.

Protections: OV lockout 1114–1170 V; 4 kV/20 us residual –> HV_BULK 1199 V; foldback 5.9 kHz; overload timer 91–189 ms –> hiccup. Rating 30 W continuous, 38 W <= 91 ms (SYS rev E 27.8 W, 30.15 W <= 200 s). V_DS 1323 / 1428 V at 1000 / 1100 V.

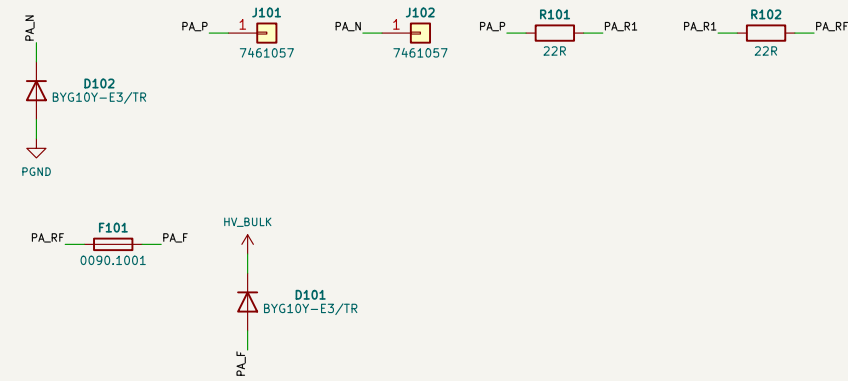
LAYOUT (not dimensions): reinforced HV <–> PELV for 1000 V DC, PD2, OVC II (IEC 62477–1 / 62109–1, ECO–10); only T1 and the CNY65B cross; 1000 V spacing between HV nets and between the ports; >= 4 cm2 2–oz Cu per SMCJ70A; VDD follower <= 85 C, away from T1 / SiC / clamp (sim report sec. 4, 6, 11).

Builds: PV all fitted / DAB–D60 port B not fitted (DAB BOM). Not bench–validated.
Sheet 1
Schematics authored by Chinmoy Bhuyan
Barrier: T1 (PD–tested, reinforced) + CNY65B, conformally coated (PD1) – D–032

Sheet: /
File: AUX–HV.kicad_sch

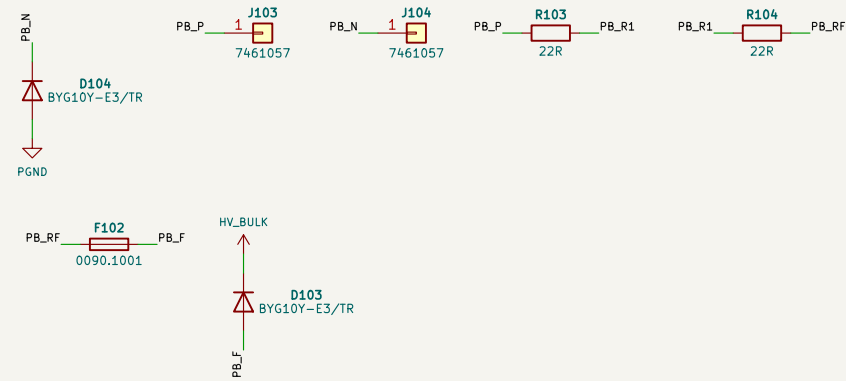
Title: AUX–HV (builds PV, DAB–D60) – Cover / Hierarchy

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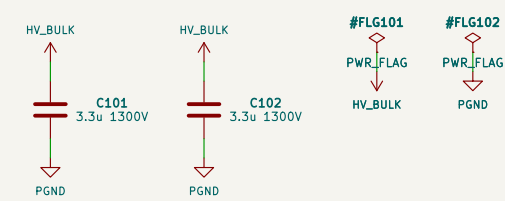
2 per port AHEAD of the fuse (AC10 pulse-rated): fuse-clearing 24 J each vs -57 J; surge 1500 V each vs 4.5 kV
Fuse: per-port fuse 1000 VDC gPV behind 44R: fault current ≤ 25 A (breaking 20 kA not needed), hot-plug I_{2t} 0.091 A_{2s} = 16 % of melting

PORT B: PV BUILD ONLY (NOT FITTED ON DAB-D60)



2 per port AHEAD of the fuse (AC10 pulse-rated): fuse-clearing 24 J each vs -57 J; surge 1500 V each vs 4.5 kV
Fuse: per-port fuse 1000 VDC gPV behind 44R: fault current ≤ 25 A (breaking 20 kA not needed), hot-plug I_{2t} 0.091 A²s = 16 % of melting

SURGE ABSORBER / INPUT FILTER



Film: 2 x film = RC surge absorber with 44R: 4 kV / 20 us residual $\rightarrow$ 1199 V (UR 1300 V); filter corner 546 Hz.

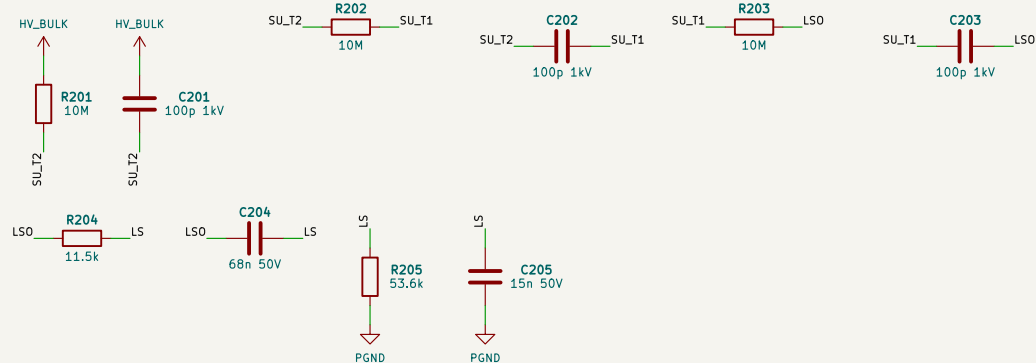
Builds: PV all fitted / DAB-D60 port B not fitted (DAB BOM). Not bench-validated.
Sheet 2
Schematics authored by Chinmoy Bhuyan
Barrier: T1 (PD-tested, reinforced) + CNY65B, conformally coated (PD1) - D-032

Sheet: /01 HV input, limit, fuse, OR/
File: 01_input.kicad_sch

Title: AUX-HV (builds PV, DAB-D60) – HV input, limit, fuse, OR

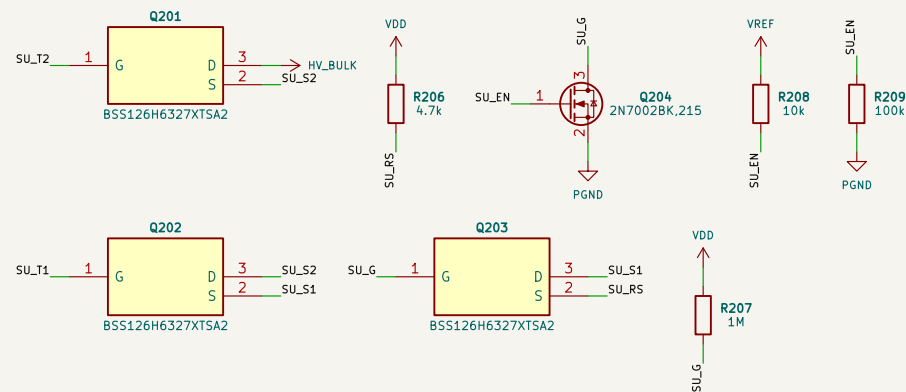
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COMPENSATED TAP STRING (ALSO THE INPUT-CAPACITOR BLEEDER)



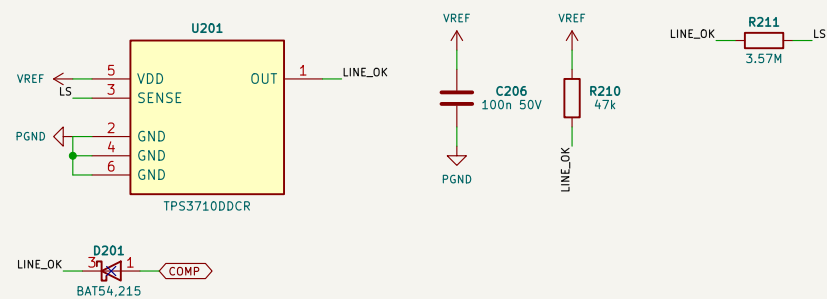
3 sections: start-up gate taps + line sense + OV tap; 367 V each at 1100 V (CRHV2512 3 kV)
string compensation (tau 1.0 ms per section)
Bottom: string bottom L50-L5: OV lockout trip 1142 V (1114-1170), release 1119 V / string bottom L5-PGND: brown-in 228 V (223-232), brown-out 183 V (175-190).

DEPLETION CASCODE CURRENT SOURCE



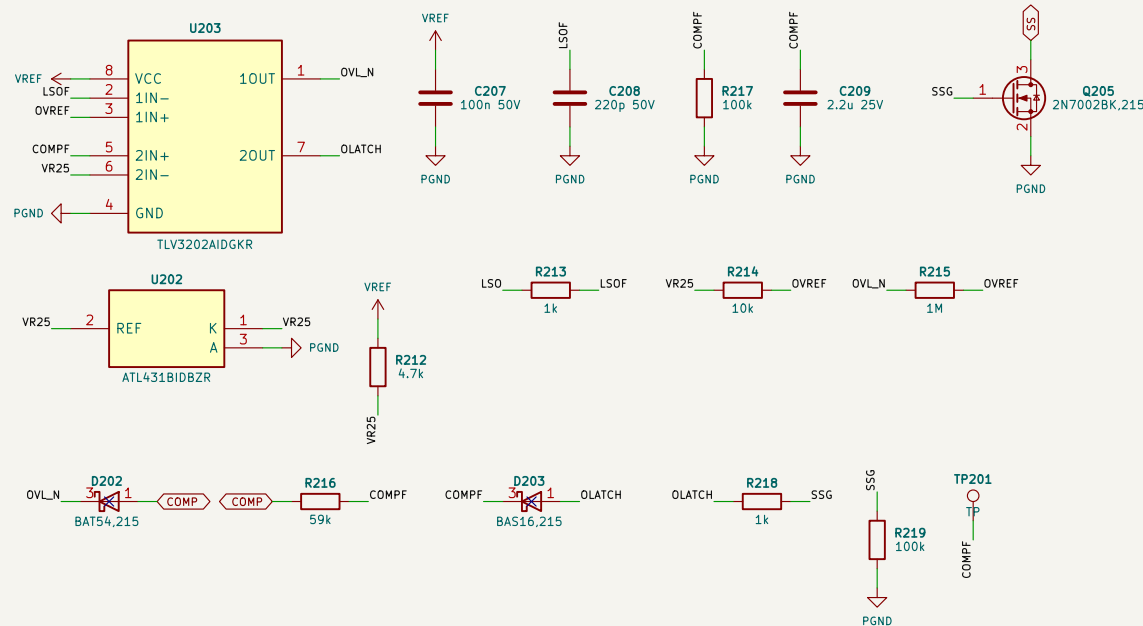
Each FET holds $-V_{in}/3$ (367 V at 1100 V). Bottom FET: $V_{GS} = -I \times R_{S_}$
start-up current 0.27–0.43 mA. V_{REF} high $\rightarrow$ Q pulls SU_G to PGND $\rightarrow V_{GS} = -V_{DD} \rightarrow$ off.

LINE BROWN-IN



LINE_OK low → BAT54 holds COMP below the 1.15 V offset (zero duty).

INPUT OV LOCKOUT AND OVERLOAD TIMER (TLV3202 ON VREF, 2.5 V REFERENCE)



Ch 1: string bottom LSO-LS: OV lockout trip 1142 V (1114-1170), release 1119 V; VOUT low pulls COMP low through a BAT54.
Ch 2: overload timer from COMP: trips when COMP > 3.89-4.06 V (34-88 W out); overload timer: 91-189 ms at full COMP. VOUT latches through a BAS16 and turns the 2N7002 on SS on until VDD reaches UVLO (VREF off resets the latch) -> hiccup.

Builds: PV all fitted / DAB-D60 port B not fitted (DAB BOM). Not bench-validated.
Sheet 3
Schematics authored by Chinmoy Bhuyan
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Sheet: /02 Start-up, brown-in, OV, timer/
File: 02_startup.kicad_sch

Title: AUX-HV (builds PV, DAB-D60) – Start-up, brown-in, OV, timer

Size: A3

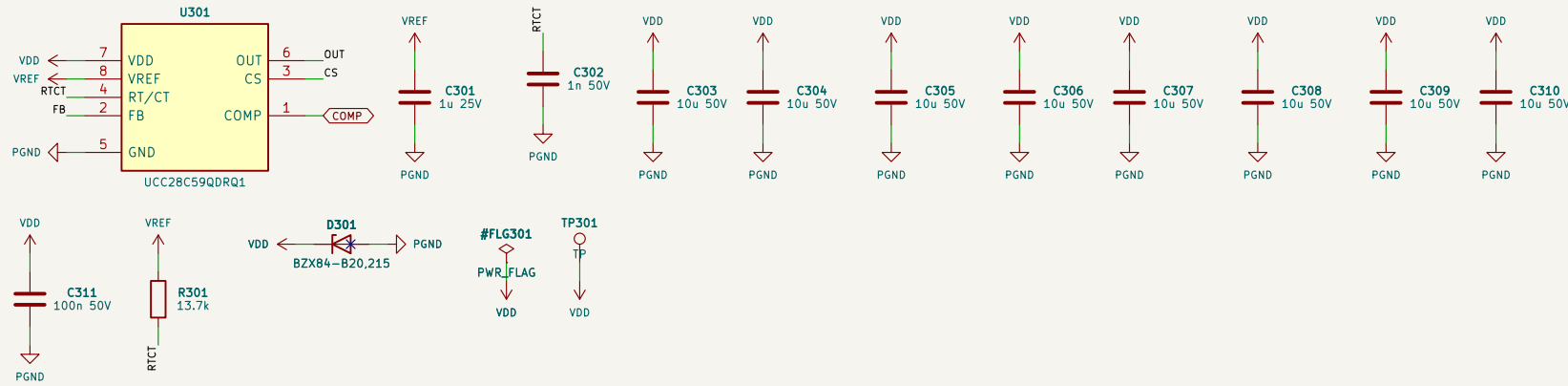
Date: 2026-10-04

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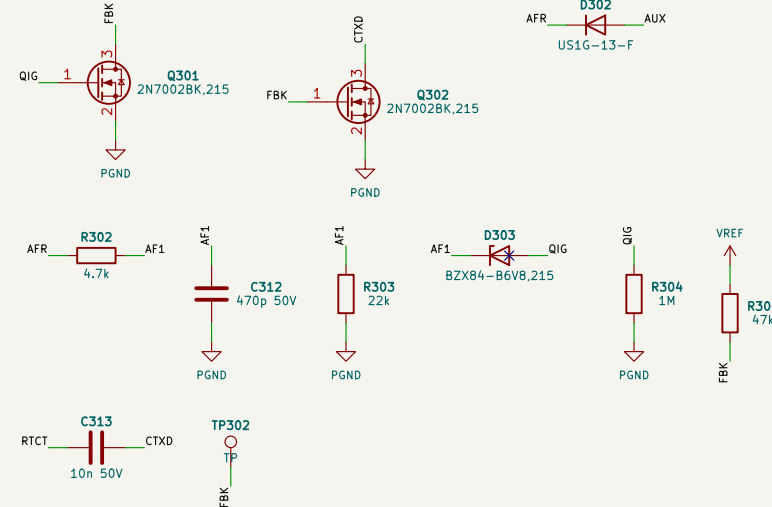
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PWM CONTROLLER



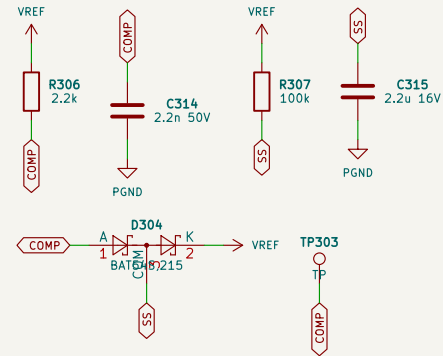
oscillator 129.8 kHz -> fsw 64.9 kHz (58.3-74.2). VDD 56 uF effective: VDD hold-up 17.6 ms. UVLO 16/12.5 V (SiC). duty < 48 %, no slope compensation needed (DCM, D < 0.5).

FREQUENCY FOLDBACK (AX-03)



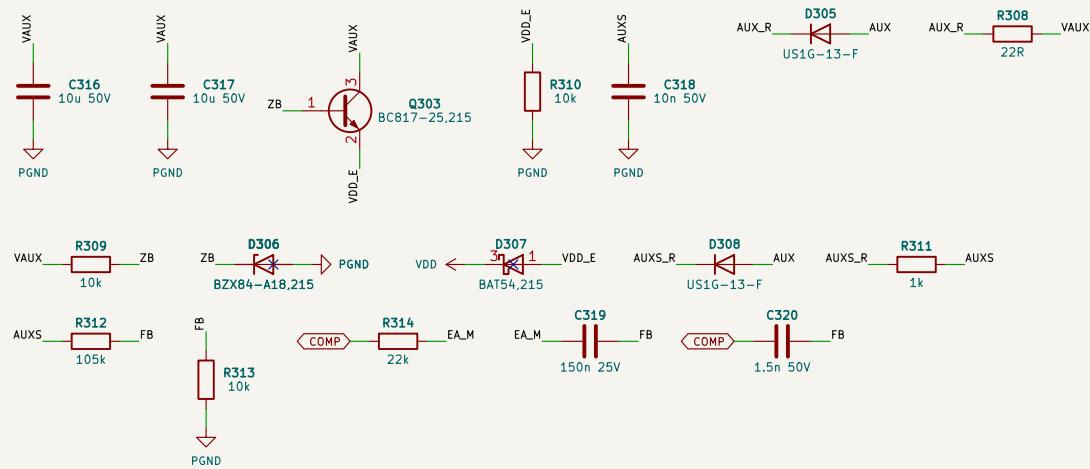
foldback: switched onto RT/CT below 7.7-10.1 V output -> fsw 5.9 kHz (5.3-6.8).
Aux plateau -> US1G -> 4.7k -> 470p || 22k -> BZX84-B6V8 -> Q_INV: aux healthy holds FBK low;
short: FBK -> VREF via 47k, Q_FB connects C_TX. aux-plateau detector charge (tau 1.8 us. rejects the clamp spike).

SOFT START AND COMP

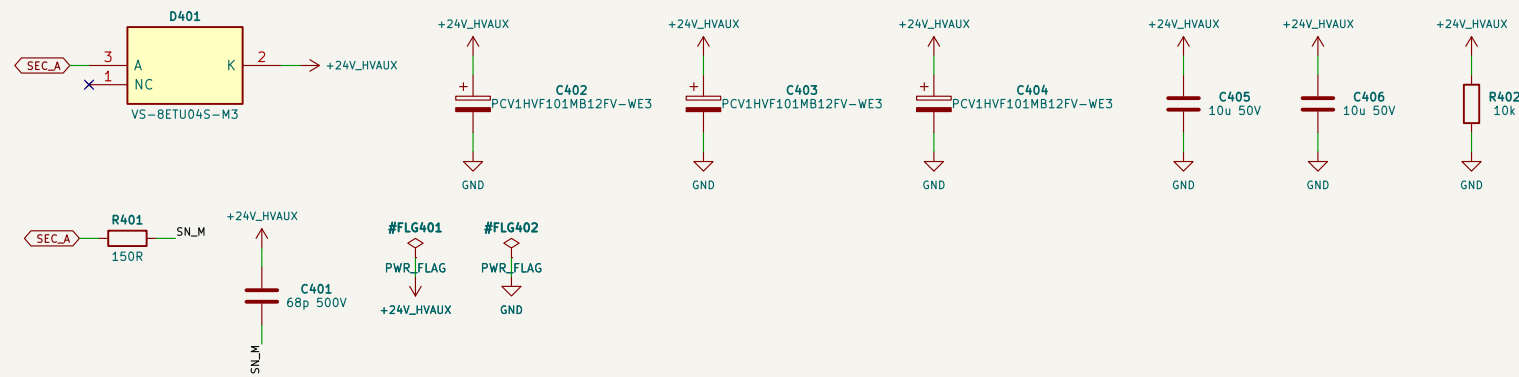


COMP = opto collector + 2.2k pull-up + EA source: BAT54S clamps COMP to the soft-start node (100k, 2.2u) and discharges it into VREF in UVLO.

AUX WINDING: VDD FOLLOWER, BACKUP OVER-VOLTAGE LOOP

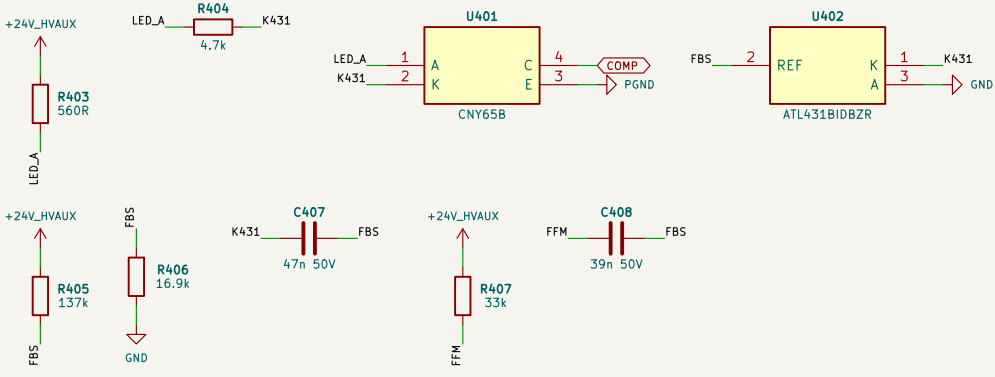


RECTIFIER AND OUTPUT CAPACITORS



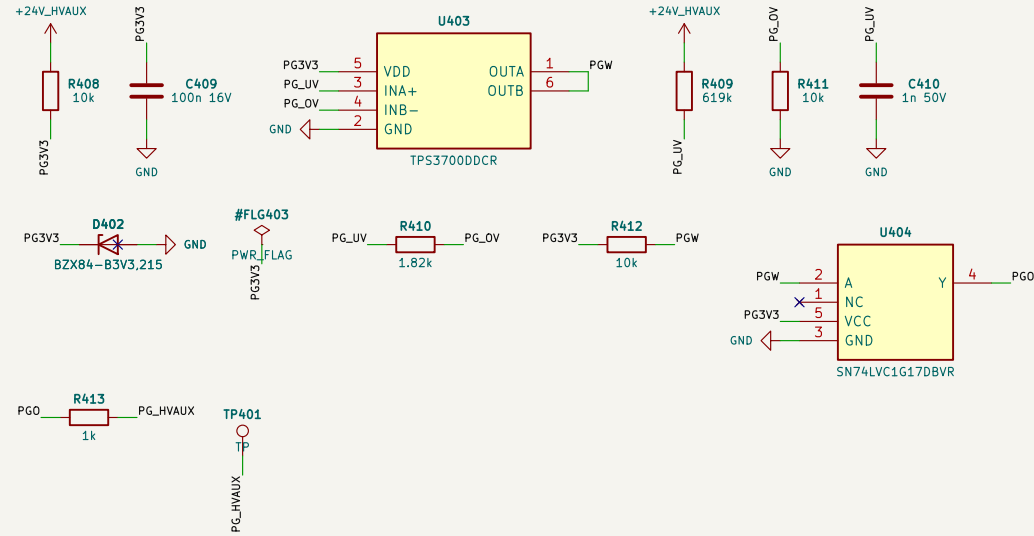
D2PAK tab = cathode on the quiet output node. Snubber 150R + 68p.
polymer bulk: 80 mV pk-pk ripple at 600 V (calc), 0.60 A rms each. Output: Vout set point 22.770 V nominal, 22.40–23.15 V.

FEEDBACK ACROSS THE BARRIER



ATL431 integrates (C_Z 47n); fast-lane: LED current up to 10.1 mA needed (R <= 1.74k), 60 mW at that current.
feed-forward to REF: no start-up / hand-over overshoot (max 23.20 V vs SYS OV 23.97 V) 33k + 39n. CNY65B collector pulls COMP on the primary.

PG_HVAUX, DRIVEN ACTIVELY HIGH (INTERFACES.AUX, INT-13)



PG window: UV falling 20.61–21.39 V, OV rising 24.93–25.54 V.
TPS3700 on a 3.3 V Zener rail; OUTA/OUTB low outside the window; SN74LVC1G17 (loff) drives the line;
PG_HVAUX series resistor (high 2.51–2.77 V into the SYS 100k pull-down). Dead or unplugged -> unpowered buffer -> the SYS-I0-AUX 100k pull-down reads not-good.

OUTPUT CONNECTOR (INTERFACES.AUX)



1 +24V_HVAUX, 2 GND, 3 PG_HVAUX – mates SYS-I0-AUX feed 3 (J103).

Builds: PV all fitted / DAB-D60 port B not fitted (DAB BOM). Not bench-validated.
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Sheet: /04 Rectifier, output, feedback, PG/
File: 04_secondary.kicad_sch

Title: AUX-HV (builds PV, DAB-D60) – Rectifier, output, feedback, PG

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